

TÜİK Happiness Analysis by Education Level

Applied Statistics Project GitHub-Compatible Documentation and Technical Report

1. Project Overview

Based on the official household survey datasets published by the Turkish Statistical Institute (TÜİK) spanning the period **2004–2022**, this project statistically investigates individuals' reported emotional states (*Happy*, *Moderate*, *Unhappy*) across varied educational attainments and models the empirical effect of education level on subjective well-being.

Core Analytical Objectives

- Benchmark happiness proportions across Higher Education, High School, Primary/Middle School, and Did Not Complete School cohorts.
- Evaluate longitudinal shifts in emotional distributions using both parametric and non-parametric hypothesis tests.
- Quantify the explanatory power of distinct education tiers on aggregate happiness scores via Multiple Linear Regression.

Employed R Libraries

Package	Domain Scope	Primary Functionality in Project
readxl	Data Ingestion	Direct ingestion of multi-sheet Excel survey data files (<code>read_excel()</code>).
tidyverse	Data Wrangling & Viz	Pipelined data transformation and grammar-of-graphics charting (<code>dplyr</code> , <code>ggplot2</code>).
knitr / kableExtra	Documentation	Reproducible document compilation and structural markdown table styling.
flextable	Reporting Formats	Dynamic generation of formatted cross-tabular previews (<code>head()</code> / <code>tail()</code>).
car / ggpubr	Statistical Diagnostics	Regression diagnostics, Levene's test, Q-Q plots, and empirical normality checks.

2. Data Ingestion and Preprocessing

The raw dataset was imported into the R computational environment via `readxl::read_excel()`. Variable data types were audited, column vectors inspected, and qualitative categorical states formatted into proper factors prior to statistical estimation.

```
# Data Ingestion projeson <- read_excel("projeson.xlsx") # Inspecting boundary records
using native forward-pipe operators projeson |> head() |> flextable() projeson |> tail() |>
flextable() # Factor transformation of categorical emotional status projeson$`Duygu Durumu`
<- as.factor(projeson$`Duygu Durumu`) str(projeson)
```

Data Pipeline Verification: All structural variables (Cohort Year, Education Level, and Emotional State Frequencies) were validated with zero missing records ($N_{\{NA\}} = 0$), standardizing proportional distributions between 0.0 and 1.0.

3. Descriptive Statistics and Exploratory Visualization

Measures of central tendency (mean, median) and dispersion (standard deviation, interquartile ranges) were computed across each educational tier to capture structural shifts in reported well-being:

```
# Central Tendency & Dispersion Measures (Higher Education Cohort)
mean(projeson$Yuksekogretim); median(projeson$Yuksekogretim); sd(projeson$Yuksekogretim) #
Comprehensive Quantile Distribution (Did Not Complete School Cohort)
summary(projeson$Yuksekogretim); quantile(projeson$`Bir Okul Bitirmede`)
```

Dot Plot Distribution

Rendered via `geom_point()` to cross-compare emotional distributions between Primary/Middle School and Higher Education graduates over 2004–2022.

Boxplot Analysis

Grouped distributions via `geom_boxplot()` revealing a downward secular trend in happiness and a corresponding upward expansion in unhappiness.

Trend Projections

Constructed using base `barplot()` and `plot(..., type="o")` to track longitudinal trajectories across census checkpoints.

4. Inferential Statistics and Hypothesis Testing

4.1. Normality Assessment

Normality assumptions for Higher Education and related cohorts were tested using Density Plots, Quantile-Quantile (Q-Q) plots, and the Shapiro-Wilk test:

- H_0 : The data follows a normal distribution.
- H_1 : The data does not follow a normal distribution.

```
# Shapiro-Wilk Normality Test (Higher Education, 2004 Baseline) Yuksekogretim_2004 <-
projeson |> filter(Yil == 2004) |> select(Yuksekogretim)
shapiro.test(Yuksekogretim_2004$Yuksekogretim)
```

Normality Test Result: The Shapiro-Wilk test yielded $p < 0.05$, leading to the rejection of H_0 . While formal parametric normality was violated, standard Student's T-Tests were applied alongside non-parametric diagnostics for comparative evaluation.

4.2. Two-Sample Mean Comparison (Student's T-Test)

Mean emotional state indices of Higher Education graduates were compared between baseline (2004) and terminal (2022) survey iterations ($H_0: \mu_{2004} = \mu_{2022}$ vs. $H_1: \mu_{2004} \neq \mu_{2022}$):

```
# Two-Sample T-Test under equal variance specification t.test(x =
projeson$Yuksekogretim[projeson$Yil == 2004], y = projeson$Yuksekogretim[projeson$Yil ==
2022], alternative = "two.sided", var.equal = TRUE)
```

Result & Inference: With $p > 0.05$, the test failed to reject H_0 . At the $\alpha = 0.05$ level, no statistically significant difference is observed between the mean emotional state scores of Higher Education graduates in 2004 versus 2022.

4.3. Population Proportion Testing

One-Sample Proportion Test

Evaluated whether the 2022 Higher Education happiness proportion equaled 0.5 via `prop.test()`. The test yielded a p -value of 0.0002247, leading to the rejection of H_0 .

Two-Sample Proportion Test

Hypothesis $H_1: p_{2022} > p_{2004}$ confirmed that the 2022 Higher Education happiness proportion was significantly greater than the 2004 proportion, with a p -value of 0.0002247.

5. Chi-Square (X²) Distribution and Independence Tests

Contingency analyses were conducted using Pearson's Chi-Square tests to establish dependence between educational attainment tiers and reported emotional states across multiple survey iterations:

Analysis Cohort / Subject	Test Specification	P-Value	Statistical Conclusion ($\alpha = 0.05$)
2004 Higher Ed. vs. Primary School Emotional State	X² Goodness of Fit	< 0.05	H₀ Rejected (Significant Link)
2022 Overall Education Level & Emotional State	Test of Independence	< 0.05	H₀ Rejected (Dependent)
2004 Overall Education Level & Emotional State	Test of Independence	< 0.05	H₀ Rejected (Dependent)
Longitudinal Years (2004, 2012, 2022) & Emotional State	Test of Independence (xtabs)	< 0.05	H₀ Rejected (Dependent)

6. Variance Comparison with F-Distribution

Equality of variances between "Happy" and "Unhappy" responses was assessed for Higher Education graduates ($H_0: \sigma^2_{Happy} = \sigma^2_{Unhappy}$):

```
# F-Test for Equality of Two Variances test <- var.test(x = Mutlu, y = Mutsuz, ratio = 1, alternative = "two.sided") # F = 5.6631, p-value = 0.1216
```

Evaluation: Since $p = 0.1216 > 0.05$, H_0 cannot be rejected. There is no statistically significant difference in variance between the two emotional state distributions.

7. Multiple Linear Regression Analysis

The qualitative emotional state variable was converted into a continuous composite index ($Happy = 1.0$, $Moderate = 0.5$, $Unhappy = 0.0$) and modeled against all education attainment categories:

```
# Model Formulation model <- lm(Duygu_Durumu_Sayisal ~ Yuksekogretim + Lise_ve_dengi_okul + Ilkokul_veya_ortaokul + Ilkokul + Bir_Okul_Bitirmedir, data = veri) summary(model)
```

0.9874	141.5	2.84e-08	
R-SQUARED (R²)	F-STATISTIC	MODEL P-VALUE	
Predictor Variable	Coefficient (β)	P-Value	Significance Status (α = 0.05)
Higher Education (Yuksekogretim)	-0.0157	0.0445	Statistically Significant
High School & Equivalent (Lise ve dengi okul)	+0.0031	0.7275	Not Significant
Primary or Middle School (Ilkokul veya ortaokul)	+0.0042	0.7178	Not Significant
Primary School (Ilkokul)	-0.0211	0.0211	Statistically Significant